Digital Forensics

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Digital Forensics (A14853)

Short Title: Digital Forensics
Faculty Science and Computing
Department Computing and Mathematics
Cluster Forensics and Security
Author JSHEPPARD
Credits 5 **Level:** Introductory

Description of Module / Aims

The aims of this module are to provide students with an understanding of digital forensics. Students will become familiar with the scientific forensic process and will apply these to the FAT file system and to online environments. Students will examine how a computer file system organizes and stores data. The student will use tools to extract and forensically analyse these kinds of data. Students will also be equipped with the skills to respond to computer attacks and to uncover information that is hidden in electronic mail messages, web pages and web servers and network communications.

Programmes

`COMP-0534`	Diploma in Computing with Security and Forensics (`WD_BCSEC_SP`)	3 / 2 / M

Indicative Content

- Common Tasks: Evidence identification, collection, analysis and presentation
- Hard Disk Acquisition: Host Protected Areas (HPAs); write blockers; cryptographic hashes
- FAT File System Analysis: File system, content, filename, metadata
- Network Based Evidence: Live response, network data, collection and analysis of data
- Web Forensics: HTTP headers; cookies; browser/server log analysis; capturing web pages; server-side data; web activity reconstruction; email headers; tracing email online

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate an awareness for issues surrounding the creation of forensic duplicates.
2. Examine FAT filesystems using a forensic toolkit.
3. Demonstrate how to retrieve deleted data.
4. Examine different types of network data that can be collected in online and off-line environments.
5. Use appropriate tools to identify, analyse and explain various network communications.
6. Reconstruct web based activity.
7. Trace e-mail.

Learning and Teaching Methods

- This module will be presented by a combination of lectures and practicals.
- The lectures will be used to introduce new topics and their related concepts.
- The practical element allows the student to put into practice the theoretical concepts covered in the lectures.
- Practical work will include the student working with standard forensic tools.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	50%	1,2,3
Assignment	50%	4,5,6,7

Assessment Criteria

<40%: Unable to describe concepts of file systems. Unable to use and explain system forensics tools and related OS utilities. Unable to acquire or interpret network data.

40%-49%: Limited knowledge and interpretation of filesystem, network and web related data.

50%-59%: Ability to discuss key concepts of the specific knowledge domain. Can use tools to moderate level, gather some data and produce some findings.

60%-69%: In addition, can interpret evidence extracted during a forensic investigation and corroborate it with other sources of evidence.

70%-100%: All above to an excellent level. Be able to discuss their effect on forensic examinations.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Lab	36	
Independent Learning	87	

Essential Material

- "forensicfocus." www.forensicfocus.com

Requested Resources

- COMPUTER LAB: BYOD Lab